QPCODE Reg.No:.....

Name:.....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) THIRD SEMESTER PRACTICAL EXAMINATION NOVEMBER 2025

DSE - MG3DSEPHY200- Basic Semiconductor Physics

(2024 ADMISSION ONWARDS)

Duration: 2 Hours Maximum Marks: 35

Record: 5 Marks

Attempt the question marked 'X'. Change of question is not allowed (30 marks)

1. Study the static and dynamic characteristics of a Diode [S, A, An] [CO 2]
2. Study the static and dynamic characteristics of a Zener Diode in the reverse bias condition. [S, A, An] [CO 2]
3. Construct a voltage regulator using a Zener diode and study line and load regulations. [S, A, An] [CO 2]
4. Construct a half wave rectifier and find the ripple factor with and without filter circuit. [S, A, An] [CO 3]
5. Construct a full wave (center tap) rectifier and find the ripple factor with and without filter circuit. [S, A, An] [CO 3]
6. Construct a full wave (bridge) rectifier and find the ripple factor with and without filter circuit. [S, A, An] [CO 3]
7. Construct a positive, negative and biased clipper circuit and study the output wave forms. [S, A, An] [CO 3]
8. Construct a positive, negative and biased clamper circuit and study the output wave forms. [S, A, An] [CO 3]
9. Construct a voltage doubler and tripler circuit. Compare their output voltage with the theoretical values. [S, A, An] [CO 2]
10. Construct a common emitter amplifier and study its amplification factor [S, A, An] [CO 5]
11. Construct an LC oscillator (Hartley/Colpitt's) using transistor and find its frequency. [S, A, An] [CO 6]